



Small-Scale Livestock Field Manual

Chickens, ducks, quail, rabbits, goats, pigs, turkeys, and bees

\$18

Open-pollinated • Community-grown • Food sovereignty for everyone

Small-Scale Livestock Field Manual

Seedwarden | \$18

Livestock changes the food sovereignty equation in one fundamental way: animals convert inputs you can grow or forage — grass, kitchen scraps, insects, grain — into protein, fat, and calories that store on the hoof until you need them. A backyard flock of chickens can produce eggs every day of the year. Rabbits produce 200+ pounds of meat annually in 24 square feet. A pair of goats can supply a household with dairy while clearing brush.

This guide covers ten animals practical for small-scale, backyard, and homestead production — not factory farming. The focus is on what an individual can manage without hired labor, specialized veterinary infrastructure, or large acreage. Every number here — space requirements, feed consumption, production output — reflects real-world small-scale experience, not commercial benchmarks.

Read your local zoning ordinances before acquiring any livestock. Regulations on backyard chickens, roosters, goats, and pigs vary enormously by municipality. What's legal in a rural county may be prohibited in a suburban lot. Check first.

Important Notice

Before acquiring any animal, check your local zoning ordinances — what is legal varies enormously by municipality, and violations can result in fines or forced removal of your animals. This guide is educational, not veterinary advice. For any animal health issue — illness, injury, abnormal behavior, or questions about medications — consult a licensed veterinarian. Livestock keeping involves real responsibilities to living creatures, and getting it wrong has consequences for the animals first and for you second.

Part One: Infrastructure Before Animals

Every livestock operation needs these fundamentals before the first animal arrives. Infrastructure failures — inadequate housing, poor fencing, no water access — account for most beginner losses.

Fencing

For chickens and ducks: Hardware cloth (welded wire, ½-inch mesh) is significantly more predator-resistant than chicken wire. Chicken wire keeps chickens in but does not stop raccoons, foxes, or dogs from tearing through it. Bury hardware cloth 12 inches underground and bend it outward at the base to prevent digging predators. Minimum 4-foot height for flightless ducks; 6 feet for chickens.

For rabbits: Hutch-based systems avoid fencing requirements entirely. Ground-level movable tractors (A-frame pens on grass) allow rotational grazing without permanent fencing.

For goats: Goats are escape artists. Electric fence (3–5 strands) or welded wire panels (minimum 48 inches tall) are required. Goats will test every weak point in any fence. They will also stand on and destroy anything they can climb. Remove anything climbable from the fence line.

For pigs: Electric fence is standard. Pigs respect a single hot wire at nose height (approximately 6–8 inches off the ground for small pigs; 12–18 inches for adults). They will root under any non-electrified fence. Two-strand electric — one at nose height, one at shoulder — is highly effective.

For all livestock: Any fence is only as good as its predator protection at night. Even in suburban areas, raccoons, foxes, coyotes, hawks, and owls will find and exploit housing vulnerabilities. Lock animals in secured housing every night.

Water Systems

All livestock require clean, unfrozen water at all times. Water deprivation causes faster productivity loss and faster death than feed deprivation.

- Chickens: $\frac{1}{2}$ pint per bird per day minimum; up to 1 pint in summer heat. Use nipple waterers for cleanliness; open fonts get fouled quickly.
- Rabbits: 6–10 oz per rabbit per day; significantly more in summer. Water bottles or sipper-tube crocks.
- Goats: $\frac{1}{2}$ –3 gallons per goat per day, more for lactating does.
- Pigs: 3–5 gallons per pig per day; up to 8 gallons in heat.

For winter: insulated water containers, heated bases, or twice-daily manual watering during freeze.

Feed Storage

Keep feed dry and rodent-proof. Metal trash cans with locking lids are an effective low-cost solution. Grains attract rats and mice, which attract predators and spread disease. Never store feed in paper bags long-term. Rotate stock — use oldest feed first. Most pellets and crumbles are good for 3–4 months properly stored; whole grains 6–12 months in cool, dry conditions.

Basic Processing Equipment

For poultry processing, at minimum:

- Killing cones (can be improvised from traffic cones or hardware cloth funnels)
- Large pot for scalding (minimum 5-gallon) and a heat source to reach 145–150°F
- Plucker (hand plucking is slow but works; a tub plucker attachment for a drill speeds it significantly)
- Sharp boning knife and poultry shears
- Cooler with ice for chilling carcasses immediately after processing

For rabbits:

- Gambrel or hooks for hanging
- Sharp knife for pelt separation and evisceration
- Bucket for offal

For goats and pigs: a block and tackle or tractor is needed for larger animals unless you have help. Plan for this before slaughter day.

Part Two: The Animals

Chickens — Laying Flock

Laying chickens are the most accessible livestock for any scale. They require minimal space, convert a wide range of feed inputs into a high-value daily product, and can be kept legally in most urban and suburban settings (roosters are often prohibited separately — check local ordinance).

Best Breeds for Small Flocks

Breed	Eggs/Year	Egg Color	Character	Notes
ISA Brown / Red Sex-Link	300-320	Brown	Calm, productive	Best layer per dollar of feed; commercial
Rhode Island Red	250-300	Brown	Hardy, assertive	Dual-purpose; roosters are aggressive
Australorp	250-300	Brown	Docile, cold-hardy	Australian developed; excellent layer
Leghorn (White)	280-320	White	Active, flighty	Best white egg layer; not cold-hardy
Plymouth Rock (Barred)	200-280	Brown	Calm, cold-hardy	Good dual-purpose bird
Easter Egger	200-280	Blue/green	Calm, curious	Colored eggs have market premium

Space Requirements

- Indoor (coop): 4 square feet per bird minimum; 6–8 sq ft preferred to reduce pecking and disease
- Outdoor (run): 10 square feet per bird minimum; more is always better
- Nesting boxes: 1 box per 3–4 hens; 12"×12" minimum; position lower than roosting bars
- Roosting bars: 8–10 inches of linear bar space per bird; position above nesting boxes; 18–24 inches off the floor minimum

Housing Requirements

Ventilation is the most critical factor in coop design. Ammonia from droppings accumulates in poorly ventilated coops and damages respiratory health. Ventilation openings should be near the roofline (hot, ammonia-laden air rises) and covered with hardware cloth. A coop can handle cold temperatures far better than it handles moisture and ammonia — prioritize airflow over insulation in most climates.

Minimum hardware:

- Solid roof (no leaks)
- Hardware cloth on all openings (windows, vents)
- Lockable door and pop door
- Easy-to-clean floor (dirt, concrete with deep litter, or droppings board)
- Roosts and nesting boxes as above

Feed

- Layer feed: 16–18% protein pellet or crumble; ¼ pound per bird per day
- Supplemental calcium: Oyster shell, offered free-choice, is essential once hens begin laying; low calcium causes thin-shelled eggs and leaches calcium from bones
- Grit: If birds have no access to soil, provide insoluble grit for grinding feed in the gizzard
- Supplemental protein: Mealworms, black soldier fly larvae, kitchen scraps (avoid raw potato, avocado, onion, chocolate, raw beans, citrus in quantity)
- Feed storage: 3–4 months maximum; whole grains last longer than processed pellets

Production

- Hens begin laying at 16–24 weeks depending on breed and day length
- Production peaks in the first laying year (year 1 after first molt)
- Molt: Hens stop laying and regrow feathers annually, typically in fall. Duration 8–16 weeks. Many small flock keepers cull spent hens at molt and replace with pullets.
- Production drops approximately 20% per year after peak
- Light: Hens need 14–16 hours of light to maintain full production. In winter, a 40-watt bulb on a timer to extend day length maintains production year-round.

Health Basics

- Marek's Disease vaccination: Get chicks vaccinated at hatchery; not available to administer at home. This is a standard practice — buy vaccinated chicks.
 - Coccidiosis: Common in young birds; medicated starter feed prevents it, or use unmedicated feed with careful management. Signs: bloody droppings, lethargy. Treatment: Corid (amprolium) in water.
 - External parasites: Lice and mites are common, especially in warm weather. Check for mites under wing feathers and around vent. Treat with permethrin dust or spray; treat housing simultaneously.
 - Respiratory disease: Chronic Respiratory Disease (CRD, Mycoplasma) and Infectious Bronchitis are common in backyard flocks. Symptoms: rattling breathing, nasal discharge. Biosecurity (don't mix new birds with established flock without quarantine) is the primary prevention.
 - Quarantine new birds: 30 days minimum in a separate space before introducing to established flock.
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Chickens — Meat Birds

Meat chickens are the fastest path from chick to food of any livestock. Properly managed, a batch of broilers goes from day-old chick to 5-pound dressed bird in 6–8 weeks.

Breeds

Cornish Cross (most common): A commercial hybrid. Grows from day-old to 6–8 lbs live weight in 6–8 weeks. Feed conversion ratio approximately 2:1 (2 lbs feed per 1 lb live weight). Disadvantages: prone to leg problems and heart failure if allowed to grow too large; must be processed on schedule or they develop health problems; they do not reproduce. Not a heritage or open-pollinated breed.

Freedom Ranger / Red Ranger: Slower-growing colored broiler. Market weight at 11–13 weeks, 5–6 lbs live weight. Better suited to pasture, more robust health, can reproduce (though offspring are variable). Feed conversion approximately 3:1.

Heritage Dual-Purpose (Barred Rock, Black Australorp, Delaware): True heritage breeds. Slower (16–20+ weeks to market weight, 4–6 lbs). Excellent foragers. Can reproduce true to type. Used historically for both eggs and meat.

Processing Sequence (All Poultry)

1. Withhold feed 12 hours before processing — empty crop reduces contamination risk during evisceration
 2. Kill: Killing cone holds the bird inverted, exposing the neck. Sever the jugular veins on both sides of the neck (not the spinal cord — the bird should bleed out, not die instantly). Allow 1–3 minutes to bleed fully.
 3. Scald: Immerse carcass in water at 145–150°F for 45–60 seconds, agitating constantly. Test by pulling a wing feather — it should release with minimal resistance. Over-scalding at higher temperatures tears the skin during plucking.
 4. Pluck: Remove feathers immediately after scalding. Work quickly — feathers set as the carcass cools. Work from wings and tail inward to the breast.
 5. Eviscerate:
 - Remove the head and feet (feet can be saved for broth)
 - Make a horizontal cut just below the keel bone; widen the opening
 - Insert hand; cup the organs and pull outward in one motion, keeping the intestines intact
 - Separate and save: liver (remove green gallbladder without puncturing), heart, gizzard (cut open, remove inner lining)
 - Remove the crop (at the neck end)
 - Remove oil gland at the base of the tail (it imparts off-flavor if left)
 6. Rinse thoroughly inside and out with cold water
 7. Chill immediately: Submerge in ice water; bring internal temperature below 40°F within 4 hours. Allow to rest in ice water 2–4 hours before refrigerating or freezing — this improves texture.
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Ducks

Ducks are more water-efficient than their reputation suggests — they need water deep enough to dip their heads (to clear their sinuses), not a pond. A rubber stock tank or child's pool changed regularly is adequate.

Best Breeds

Breed	Purpose	Eggs/Year	Notes
Khaki Campbell	Eggs	250–340	One of the most prolific egg layers of a
Indian Runner	Eggs	150–300	Upright posture, excellent foragers, lim
Pekin	Meat	125–225	Most common commercial meat duck; 7–8 l
Muscovy	Dual	60–120	Technically not a duck (<i>Cairina moschata</i>
Welsh Harlequin	Dual	240–330	Good eggs, decent meat, beautiful plumag

Space Requirements

- Indoor: 6 square feet per duck minimum; no roosts needed (they sleep on the floor)
- Outdoor: 15 square feet per duck minimum; they foul soil quickly — rotate pasture or use deep bedding
- Ducks do not need nesting boxes but will lay on the floor in a corner; collect eggs early morning (ducks lay at night and early morning)

Processing

Same basic process as chickens with two differences: ducks are harder to pluck (more feathers, wax-like waterproofing). Scald at **155–160°F** for 2–3 minutes, or use a wax-dipping system (dip in paraffin wax, let set, peel off with feathers). Alternatively, skin the duck entirely (faster, lose the skin).

Quail — Coturnix

Coturnix (Japanese) quail are the most space-efficient meat and egg animal available. They require approximately 1 square foot per bird in cages, begin laying eggs at 6–8 weeks, lay 250–300 eggs per year, and reach processing weight at 6–8 weeks. A colony of 20 birds in a 20-square-foot cage produces eggs nearly year-round.

Production

- Eggs: Begin at 6–8 weeks. Quail eggs are approximately 1/5 the size of a chicken egg; it takes 4–5 quail eggs to equal 1 chicken egg in volume. Quail eggs are higher in relative protein and fat.
- Meat: Process at 6–8 weeks, approximately 6–9 oz live weight, yielding a 4–6 oz dressed bird. One quail is a single large serving.
- Feed: Approximately ½ oz (14g) of feed per bird per day. A 50-lb bag of feed runs 20 birds for approximately 70 days.

Housing

Battery cage systems or colony pens with solid floors. Wire floors allow droppings to fall through (cleaner, less disease) but require soft landing zones to prevent foot injuries. Quail fly explosively when startled — cover cages to prevent head strikes. Males crow; plan accordingly in urban settings.

Processing Quail

Same process as chickens at small scale. No scalding — quail are typically skinned. Hold the bird in one hand, make a small cut through the skin at the back of the neck, peel the skin off in one motion. Eviscerate through a small abdominal cut. The entire process takes under 2 minutes per bird with practice.

Rabbits

Rabbits are the most efficient meat animal per square foot of any livestock. A single doe produces 4–8 litters per year of 6–10 kits each — potentially 50+ market-weight rabbits per doe per year, in a hutch that fits in a garage.

Best Meat Breeds

Breed	Market Weight	Days to Market	Dressed Yield
New Zealand White	5 lbs	56-70 days	~55% of live weight
California	4.5-5 lbs	56-70 days	~55%
Rex	4.5-5 lbs	70-80 days	~56%; pelt has commercial value
Flemish Giant	8-12 lbs	90-120 days	~50%; very large; more feed input

Space Requirements

- Individual hutch: Minimum 30"×36"×18" per rabbit; larger is better for welfare and growth
- Breeding trio: 1 buck, 2 does minimum for sustainable production. A 2-doe setup produces market rabbits approximately 8–10 months of the year.
- Kits: Leave with the doe in her hutch until 8 weeks, then move to a grow-out pen

Reproduction

- Age at first breeding: Does at 4–5 months (smaller breeds) to 6–7 months (larger). Buck at 5–6 months.
- Breeding: Place doe in buck's hutch (never the reverse — does are territorial). Remove after 20–30 minutes. Repeat 8 hours later. Confirm pregnancy by palpation at 10–14 days (small marble-sized embryos felt in the abdomen).
- Gestation: 30–31 days
- Kindle (birth): Doe pulls fur to line a nest box. Check next day — count kits, remove any dead. Hands-on inspection is fine; does do not routinely abandon kits due to human scent.
- Litter size: 6–10 kits; first-time does often have smaller litters
- Re-breeding: Does can be rebred 2–4 weeks after kindling. Intensive (2-week re-breed) maximizes production but stresses the doe.

Feed

- Pellets: 4–6 oz per adult rabbit per day; unlimited for does with litters and growing kits
- Hay: Timothy or orchard grass hay should be available at all times — critical for gut motility
- Fresh greens and garden scraps: Supplement freely with leafy greens, carrot tops, kale; introduce slowly to avoid GI upset. Avoid iceberg lettuce, dried beans, and anything that would cause GI distress in humans.
- Never feed: Rhubarb, potato leaves, raw beans, chocolate, avocado.

Processing Rabbits

1. **Dispatch:** Hold rabbit by rear legs; stretch the neck firmly between thumb and first two fingers of the other hand and give a firm, sharp downward pull and push simultaneously — this cervical dislocation is the standard humane method for rabbits. Alternatively, a sharp blow to the base of the skull with a heavy dowel.
2. **Hang:** Hook rear legs between tendon and bone just above the ankle; hang from a gambrel or two hooks at a workable height.
3. **Remove the head:** Cut through the neck with a sharp knife.
4. **Skin:** Make a circular cut around each rear ankle. Cut the skin from ankle to ankle across the crotch. Grip the skin at the hips and pull downward firmly — it will peel off in one motion, inside-out, down to the front legs and neck. Pull each front leg through the skin; remove.
5. **Eviscerate:** Make a shallow cut from pelvis to sternum, being careful not to puncture intestines. The viscera will fall forward. Save the liver (remove gallbladder), heart, and kidneys. Remove the bladder (at the pelvic floor) carefully.
6. **Rinse and chill immediately in cold water.**

Dressed yield: Approximately 55% of live weight. A 5-lb market rabbit produces approximately 2.75 lbs of dressed meat.

Pelt: Rex pelts especially have value if properly processed. Remove the pelt inside-out, scrape any fat off the inner surface, stretch on a wire frame, and allow to dry. Salt-cured pelts can be stored before tanning.

Dairy Goats

Goats produce milk that is more easily digestible than cow's milk, requires no large pasture, and can thrive on brush, weeds, and poor forage that few other livestock would touch. Two does in milk can supply a household with fresh milk, cheese, yogurt, and butter year-round.

Best Breeds

Breed	Milk/Day	Butterfat	Size	Notes
Nigerian Dwarf	1-2 quarts	6-10%	40-60 lbs	Highest butterfat; ideal for small spaces
Nubian	2-3 quarts	4-5%	100-150 lbs	Rich milk; pendulous ears; vocal
Alpine	2-4 quarts	3-4%	100-130 lbs	High production; hardy
LaMancha	2-4 quarts	3.5-4.5%	100-130 lbs	Excellent producer; tiny ears ("gopher ears")
Oberhasli	2-3 quarts	3.5-4%	90-120 lbs	Quiet, calm; bay coloring

Space Requirements

- Indoor housing: 15–20 square feet per goat in a shared barn or shelter
- Outdoor: 200 square feet per goat minimum; more is better for health and foraging
- Fencing: As noted above — welded wire panels at minimum 48 inches, or 3-strand electric

Reproduction and Freshening

- Breeding season: Most breeds are seasonally polyestrous (breed fall through early winter). Nigerian Dwarfs are year-round breeders.
- Estrus cycle: 19–21 days; signs include tail flagging, vocalization, mucus discharge, reduced milk production
- Gestation: 150 days (5 months)
- Kids: 1–3 per doe is typical; triplets are common in high-producing breeds. Kids need colostrum within the first hour of life.
- Freshening: A doe comes into milk after kidding and remains in milk for 9–10 months if milked consistently. Dry off the doe 2 months before the next kidding to allow the udder to rest.

Milking

- Frequency: Twice daily, 12 hours apart, is standard for maximum production. Once daily reduces production approximately 20–25% but is more manageable for small operations.
- Hygiene: Clean the udder with a teat dip or warm cloth before milking. Strip the first few streams from each teat into a strip cup (checks for mastitis — clumps indicate infection). Milk into a clean, sanitized container. Chill milk immediately to below 40°F.
- Milking stand: A stand that holds the goat's head in a stanchion while she eats makes milking solo practical and safe.

Feed for Dairy Goats

- Hay: Good-quality grass hay (timothy, orchard grass, bermuda) should be available free-choice. Alfalfa provides additional protein and calcium and is preferred for lactating does.
- Grain: Dairy does in milk need grain to maintain body condition. Feed 1 lb of 16% goat grain per 3 lbs of milk produced daily. A doe producing 2 quarts (approximately 4 lbs) of milk gets approximately 1.3 lbs of grain per day, split between two milkings.
- Minerals: Loose goat minerals free-choice. Goats are copper-sensitive if overdosed; use goat-specific minerals, not cattle minerals.
- Baking soda: Free-choice baking soda helps buffer rumen pH; important when feeding significant grain.

Common Health Issues in Goats

- CAE (Caprine Arthritis and Encephalitis): Viral; spread through infected milk to kids. Buy CAE-negative tested animals. Prevent by heat-treating colostrum (135°F for 1 hour) and feeding pasteurized milk to kids.
 - CL (Caseous Lymphadenitis): Contagious abscess disease. Avoid introducing affected animals. Do not pop abscesses — the bacteria (*Corynebacterium pseudotuberculosis*) are extremely persistent in soil.
 - Enterotoxemia (overeating disease): Caused by *Clostridium perfringens*, often triggered by sudden feed changes. Vaccinate with CD/T vaccine; booster annually; does 4–6 weeks before kidding.
 - Internal parasites (barber pole worm, *Haemonchus contortus*): The primary killer of goats in humid climates. FAMACHA scoring (checking eyelid color for anemia) guides targeted deworming. Rotate pastures. Do not deworm the whole herd routinely — this drives resistance.
 - Hoof trimming: Every 6–8 weeks; overgrown hooves cause lameness and joint damage.
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Meat Goats

Meat goats require fewer inputs than dairy goats and are more suited to rough pasture, brush, and marginal land. They require no daily milking and can largely self-manage on good pasture with minimal intervention.

Breeds

Boer: The standard commercial meat goat. Fast growth, heavy muscling. Market weight 60–80 lbs at 3–5 months. Buck can reach 200–300 lbs. Susceptible to internal parasites in humid climates.

Kiko: Developed in New Zealand for parasite resistance and hardiness. Slower growth than Boer but significantly lower health inputs. Excellent choice for humid southeastern US.

Spanish Goat: Traditional American landrace. Extremely hardy and parasite-resistant. Slower growth; more suited to range management than intensive production.

Boer × Kiko or Boer × Spanish crosses offer improved parasite resistance with better growth rates than pure Kiko or Spanish.

Processing

Goats are typically slaughtered by a mobile processor or at a USDA-inspected facility (for retail sale). For home use, the process is similar to sheep: stun or cut the carotid arteries and jugular, bleed out, skin starting at the legs, remove viscera, split the carcass, age at 35–40°F for 5–7 days. Market weight of 60–80 lbs live yields approximately **35–45 lbs hanging weight** (45–55% yield).

Pigs

A single pig raised from weaner (8 weeks old, approximately 40 lbs) to market weight (250–280 lbs) in 5–6 months provides approximately 180–200 lbs of pork in the freezer. Two pigs together grow faster and more contentedly than one — pigs are social and become stressed alone.

Heritage Breeds for Small-Scale Production

Breed	Character	Fat Level	Notes
Berkshire	Excellent flavor, pink meat	Moderate	Top choice for flavor; black with white
Tamworth	Lean, excellent for bacon, active forager	Low	Best for pasture-based systems
Duroc	Fast growth, good temperament	Moderate	Red coloring; excellent feed efficiency
Gloucestershire Old Spot	Hardy, excellent forager	Higher	Traditional orchard pig
Red Wattle	Good temperament, flavorful	Moderate	Endangered heritage breed

Commercial crosses (Yorkshire × Duroc, etc.) grow faster than heritage breeds — market weight in 5 months vs. 6–8 months for heritage. Heritage breeds develop more intramuscular fat and better flavor.

Space Requirements

- Minimum outdoor space: 50 square feet per pig in a dry lot; ideally 200–400 sq ft for two pigs in a paddock rotation system
- Shelter: Pigs need shade and a wallowing area — they cannot sweat and die quickly in heat without cooling. A simple three-sided structure is adequate in temperate climates.
- Rooting: Pigs will root up any turf and turn a paddock into bare ground in weeks. Rotational paddock systems (moving pigs to fresh ground every 2–4 weeks) allow pasture recovery. Nose ringing reduces rooting but decreases the welfare of pasture pigs.

Feed

- Pigs eat approximately 3–4% of body weight in dry feed per day — roughly 6–8 lbs of feed for a 200-lb pig
- Commercial complete hog feed: 14–16% protein; most convenient; expensive at scale
- Grain-based ration: Corn or barley (energy source) + soybean meal (protein) mixed approximately 4:1. Supplement with a hog mineral pack.
- Kitchen and garden scraps: Pigs can utilize most food waste, dairy products, garden trimmings, stale bread. Many municipalities prohibit feeding garbage to pigs — check local regulations. Do not feed raw meat to pigs (disease vectors).
- Pasture: Provides nutrients, minerals, and foraging enrichment; not sufficient as a sole feed source for fast growth
- Milk and whey: If you have a dairy operation, excess milk and whey are excellent high-protein pig feed

Feed-to-gain ratio: Commercial pigs convert approximately 2.5–3 lbs of feed per 1 lb of gain. Heritage breeds on pasture: 3.5–4.5 lbs feed per lb gain.

Slaughter and Yield

Slaughter at **230–280 lbs live weight**. At this weight, a pig yields approximately:

- Hanging weight: 72–75% of live weight (approximately 170–200 lbs)
- Retail cuts: 65–70% of hanging weight (approximately 140–160 lbs of packaged pork)

A professional butcher is the most practical option for most small-scale producers. Home slaughter requires a .22 rifle or captive bolt stunner, adequate hanging infrastructure, and a scalding tank large enough for the carcass (or a skid steer with a bucket of boiling water). It is labor-intensive and requires at least two people.

All parts are useful:

- Leaf lard (kidney fat): render to the finest cooking and preservation fat
- Head: head cheese, brawn, stock
- Feet: stock, pickled pigs' feet
- Organs: liver, heart, kidneys are all edible and nutritious; prepare fresh
- Skin: cracklins (fried), chicharrones; or tan for leather
- Intestines: natural sausage casings after thorough cleaning

Turkeys

Turkeys require more management than chickens, are more vulnerable to disease (particularly blackhead disease from contact with chicken-raising areas), and take significantly longer to reach market weight. The payoff is a large, flavorful bird that produces substantially more meat per animal than any other poultry.

Breeds

Broad Breasted White / Bronze (commercial): Reach market weight in 16–20 weeks (toms 30–45 lbs, hens 15–25 lbs). Cannot reproduce naturally — the breast muscle mass prevents mating. Must be purchased as poults each season.

Heritage breeds (Narragansett, Bronze, Bourbon Red, Slate, Royal Palm): Smaller market weight (toms 20–25 lbs, hens 10–14 lbs) at 25–30 weeks. Reproduce naturally. Better flavor and significantly more active, requiring more space.

Management Notes

- Poults (young turkeys) need heat (95°F for week 1, reducing 5°F per week) and are fragile for the first 4 weeks.
- Never raise turkeys on land that has held chickens within the past 3 years — *Histomonas meleagridis* (blackhead disease) lives in the soil and is lethal to turkeys while chickens are largely carriers.
- Turkeys require higher protein feed than chickens — 28% protein starter for the first 8 weeks, transitioning to 20% for the grow-out phase.

Processing

Same as chicken; scale up for the larger bird. Scald at 150–155°F for 2–3 minutes. A larger scalding vessel (a galvanized wash tub over a propane burner is common) is needed. Heritage toms in particular have long feathers requiring more thorough scalding.

Honeybees

Bees produce honey (a preservation medium, a sweetener, and a caloric food), beeswax (candles, wood finish, skin care, grafting wax), propolis (antimicrobial resin used in natural medicine), and pollination services for the rest of your growing operation. A healthy hive can produce **40–80 lbs of surplus honey per year** in a good nectar season.

Getting Started

Langstroth hives are the standard in North America — stackable boxes with removable frames that allow inspection without destroying comb. A starter kit typically includes:

- 2 deep hive boxes (brood chambers)
- 1–2 medium honey supers
- Frames with wax or plastic foundation
- Inner cover and telescoping outer cover
- Bottom board (screened is preferred for varroa mite management)

Essential tools:

- Smoker (pine straw, burlap, wood chips — fuel that produces cool, white smoke)
- Hive tool (flat and hooked for prying frames)
- Full veil or jacket with attached veil at minimum
- Gloves (leather or thick rubber)

Starting a colony:

- Package bees: 3 lbs of bees + a caged queen; install in spring when temperatures are reliably above 55°F
- Nucleus colony (nuc): 5 frames of established brood, bees, and a laying queen; faster establishment, more expensive
- Swarm capture: A free colony; requires knowing where swarms congregate locally and being available to collect them

Basic Management Calendar

- Spring: First inspection of the year; assess brood pattern and queen health; add supers when nectar flow begins; check for swarm preparations
- Summer: Add honey supers as needed; check for swarm cells; early summer is the highest production period in most regions
- Fall: Extract honey after main nectar flow ends; consolidate to 2 deep boxes; treat for Varroa mites; ensure adequate winter stores (60–80 lbs of honey for northern climates)
- Winter: Minimal intervention; check for food stores in late winter; provide emergency fondant if colony is light

Varroa Mite Management

Varroa destructor is the single greatest threat to managed honeybee colonies. Untreated colonies typically collapse within 2–3 years. Treatment options:

- Oxalic acid (Apivar): Approved organic treatment; highly effective; apply in winter when brood is minimal
- Formic acid (Mite Away Quick Strips): Effective; works on mites in capped cells; temperature-sensitive
- Apiguard (thymol): Temperature-dependent; best for fall treatment

No treatment is not a sustainable management strategy in most of North America.

Honey Extraction

1. Remove supers when frames are 80%+ capped with wax
2. Uncap cells with an uncapping knife or fork
3. Spin in a radial or tangential extractor (hand-cranked extractors are available)
4. Strain through 400-micron and 200-micron filters
5. Allow to settle 24–48 hours; bottle

Raw honey stored in capped cells is shelf-stable indefinitely at low humidity. Extracted honey sealed in jars keeps 1–2 years before moisture absorption becomes an issue.

Part Three: Multi-Animal Integration

Well-designed multi-animal systems reduce inputs, improve soil fertility, and increase productivity per acre.

Chickens following cattle or pigs: Chickens scratch through manure and eat fly larvae, parasites, and undigested grain. Reduces fly pressure and parasite load for the larger animal; provides enrichment and additional nutrition for chickens.

Goats and chickens: Compatible; goats eat browse and chickens eat ground-level feed. Goat manure supports insect life that chickens forage. Separate feeding areas prevent goats from consuming chicken feed (chickens do not tolerate goat minerals).

Ducks and garden: Ducks excel at slug and snail control in the garden. They are messy around water and will eat young transplants — better for cleared beds or orchard areas than active vegetable rows.

Pigs as land clearers: Pigs will clear scrub brush, root up stumps, and break down heavy sod. One pig per 1,000 sq ft for 4–6 weeks will clear and till ground better than any mechanical implement. Rotate chickens through afterward to further scratch and seed.

Rabbits and compost: Rabbit manure is "cold" (unlike pig or chicken manure) and can go directly onto garden beds without composting. Worm beds under rabbit hutches convert manure directly into vermicompost.

Part Four: Growing Your Own Feed

Purchasing commercial feed is convenient but creates dependency. These crops can be grown on-site to supplement or partially replace purchased feed:

Crop	Primary use	Animal
Field corn / dent corn	Energy grain	All species
Sunflower seeds	Oil/protein supplement	Chickens, ducks
Amaranth	Protein-rich grain/green	Chickens, rabbits
Sorghum / milo	Energy grain, heat-tolerant	Chickens

Fodder beets / mangel-wurzels	Calorie-dense root	Pigs, goats, chickens
Buckwheat	Fast-growing protein	Chickens, pigs
Comfrey	High-protein leafy green	Rabbits, pigs, goats
Black soldier fly larvae	High-protein feed	Chickens, fish, pigs
Duckweed	Aquatic high-protein	Chickens, ducks, fish
Alfalfa / clover	Protein-rich hay/pasture	Goats, rabbits

A modest 1/8-acre plot planted intensively in field corn, sunflowers, and fodder beets can meaningfully reduce purchased feed costs for a small flock of chickens or rabbits.

Part Five: Legal Considerations

Zoning laws for backyard livestock vary dramatically. Before acquiring any animal:

Common urban/suburban restrictions:

- Roosters are prohibited in most municipalities regardless of chicken allowances (noise)
- Chicken limits typically range from 4–12 hens; some jurisdictions allow up to 25
- Goats: dwarf or miniature breeds are sometimes permitted where full-size goats are not; dairy goats are more commonly permitted than meat production
- Pigs: prohibited in most urban and suburban zones; some areas permit "miniature" pigs
- Rabbits: often treated as pets rather than livestock; fewer restrictions in most areas
- Bees: increasingly permitted in urban areas; setback requirements from property lines vary (typically 10–25 feet from property boundaries); water source requirement is common

Agricultural exemptions: Many rural counties have right-to-farm protections and minimal livestock regulation. Check county ordinances, not just municipal ordinances — they may differ.

Water and waste: In many jurisdictions, livestock operations require a waste management plan. Composting manure on-site and using it as garden amendment is the most common small-scale solution.

Quick Reference — Production Summary

Animal	Space (min.)	Feed/Day	Start Age	Production	Time to Market
Laying hens	4 sq ft/bird indoor	¼ lb/bird	16–24 wks	5–6 eggs/hen/week	N/A (egg production)
Meat chickens (Cornish)	4 sq ft/bird	0.2–0.5 lb/bird	1 day	5–6 lbs live	6–8 weeks
Ducks (Pekin, meat)	6 sq ft/bird	¼ lb/bird	1 day	7–8 lbs live	7–8 weeks
Coturnix quail	1 sq ft/bird	½ oz/bird	6–8 wks	250–300 eggs/year	6–8 weeks
Rabbits	30"×36" hutch	4–6 oz/adult	5–6 mos	6–10 kits/litter, 4–8 litters/year	8–10 weeks from kindle
Dairy goats	15 sq ft indoor	1.5–2 lbs grain + hay	1–2 yrs	1–4 qt milk/day	N/A (dairy)
Meat goats	50 sq ft	Hay + browse	3–5 mos	35–45 lbs hanging weight	7 months
Pigs	50–100 sq ft	6–8 lbs/day	8 wks (weaner)	170–200 lbs freezer weight	5–7 months
Turkeys (BBW)	6 sq ft/bird	½–1 lb/bird	1 day	25–40 lbs live (tom)	16–20 weeks

Honeybees	1 Langstroth hive	N/A (forage)	Spring nuc	40–80 lbs honey/year	N/A
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Related Seedwarden guides:

- Meat, Fish & Animal Products Preservation Field Manual – how to preserve everything these animals produce
- Harvest Preservation Field Manual – preserving the plant feed crops you grow
- Survival Garden Regional Plans – what to grow alongside your livestock operation

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If you found this guide useful, you might also like:

- Hunting, Fishing & Trapping Field Manual (\$20) – Wild-harvest skills for supplementing your homestead protein
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